CARBOplatin 70mg/m^2 and 5-Fluorouracil 600mg/m^2 with Radiotherapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved Reimbursement Status*
Locally Advanced Squamous Cell Carcinoma of the Head and Neck for patients ineligible for CISplatin based treatment	C00-C14, C30-C32,C76	00589a	N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

CARBOplatin and 5-Fluorouracil are administered on Day 1-4 every 21 days for three cycles.
Chemotherapy is only to be administered if concurrent with radiotherapy.
Facilities to treat anaphylasis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Admin. Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1-4	CARBOplatin	70mg/m^2	IV Infusion	250mL glucose 5% over 30 minutes	Every 21 days for three cycles
2	1-4	5-Fluorouracil[a]	600mg/m^2/day	Continuous IV infusion	[a]Over 96 hours in 0.9% NaCl	Every 21 days for three cycles
[a]Alternatively can be administered at 600mg/m^2 in 1000mL 0.9% NaCl as an intravenous infusion over 22 hours on days 1-4 for a total dose of 2400mg/m2 over 96 hours.						
[b]See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency						

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

ELIGIBILITY:

- Indication as above
- Stage III or IV squamous cell carcinoma of the oropharynx without evidence of distant metastases
- ECOG 0-2
- Criteria for platinum eligibility in Head & Neck squamous cell cancer should be made with reference to Szturz et al (Ref 3) with final determination made by the treating Consultant.

USE WITH CAUTION:

- Pre-existing motor or sensory neuropathy > grade 2

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EXCLUSIONS:

- Hypersensitivity to CARBOplatin*, 5-Fluorouracil or any of the excipients
- Neutrophil count <1.5x10^9/L
- Platelet count <100x10^9/L
- Renal insufficiency with creatinine >120 micromol/L or Creatinine clearance ≤50 mL/min
- Bilirubin >1.5xULN
- Known complete dihydropyrimidine dehydrogenase (DPD) deficiency

*If it is felt that the patient may have a major clinical benefit from CARBOplatin, it may in exceptional circumstances be feasible to rechallenge a patient with a prior mild hypersensitivity reaction e.g. using a desensitisation protocol, but only with immunology advice, premedication as advised, and a desensitisation protocol under carefully controlled conditions with resuscitation facilities available and medical and/or ITU/ HDU supervision

PRESCRIPTIVE AUTHORITY:

- The treatment plan must be initiated by a Consultant Medical Oncologist

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Audiology referral as clincailly indicated
- Isotope GFR measurement (preferred) or GFR / Cr Clearance estimation
- Dihydropyrimidine dehydrogenase (DPD) testing prior to first treatment with 5-Fluorouracil using phenotype and/or genotype testing unless patient has been previously tested
    - In patients with moderate or severe renal impairment, blood uracil levels used for dihydropyrimidine dehydrogenase (DPD) phenotyping should be interpreted with caution, as impaired kidney function can lead to increased uracil blood levels. Consequently, there is an increased risk for incorrect diagnosis of DPD deficiency, which may result in under dosing of 5-Fluorouracil or other fluoropyrimidines, leading to reduced treatment efficacy. Genotype testing for DPD deficiency should be considered for patients with renal impairment.

Regular tests:

- FBC, renal and liver profile prior to each cycle
- Assessment of hands and feet for palmar-plantar erythrodysaesthesia (PPE) as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

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DOSE MODIFICATIONS:

- Consider a reduced starting dose in patients with identified partial DPD deficiency.
  - Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring.
- Any dose modification should be discussed with a Consultant.

Haematological:

Table 1: Dose modification of CARBOplatin and 5-Fluorouracil in haematological toxicity

ANC ($x10^9$/L)		Platelets ($x10^9$/L)	Dose
≥1.5	and	≥100	100%
1.0-1.5	or	75-99	75%
<1.0	or	<75	Delay one week

Renal and Hepatic Impairment:

Table 2: Dose modification of CARBOplatin and 5-Fluorouracil in renal and hepatic impairment

Drug	Renal Impairment	Hepatic Impairment	
CARBOplatin[a]	*See below	No dose modification required	
5-Fluorouracil[b]	No need for dose adjustment is expected	Mild and moderate (without renal impairment):	No need for dose adjustment is expected
	Haemodialysis : No need for dose adjustment is expected	Severe:	Not recommended

[a]Renal – Content agreed by clinicians for use across NCCP regimens, Hepatic – Giraud et al 2023
[b]Renal and hepatic dose modifications as per Giraud et al 2023

***Renal dysfunction and CARBOplatin:**

- Patients with creatinine clearance values of < 60mL/minute are at greater risk to develop myelosuppression
- In case of GFR ≤ 20mL/minute CARBOplatin should not be administered at all
- If Cockcroft & Gault or Wright formula are used, the dose should be calculated as required per cycle based on a serum creatinine obtained within 48 hrs of drug administration
- If isotope GFR is used, the dose should remain the same provided the serum creatinine is ≤110% of its value at the time of the isotope measurement. If the serum creatinine is higher than this, consideration should be given to remeasuring the GFR or to recalculating using Cockcroft & Gault or Wright formulae

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Management of adverse events:

Table 3: Dose Modification for Adverse Events in 5-Fluorouracil

Adverse reactions	Recommended dose modification
Dysphagia or stomatitis:	
Grade 0-2	100%
Grade 3	Delay until improvement and proceed at 75-100%
Grade 4	Discontinue
Weight loss from baseline:	
≤10%	100%
>10%	75% if hyperalimentation instituted, otherwise discontinue
Palmar-plantar erythrodysaesthesia:	
Grade 0-2	100%
Grade 3	75%

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CARBOplatin: Moderate **(Refer to local policy).**
5-Fluorouracil: Low **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: No specific recommendations

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OTHER SUPPORTIVE CARE:

- Anti-diarrhoeal treatment **(Refer to local policy).**
- Management of PPE (refer to local policy)

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics for details.

REGIMEN SPECIFIC COMPLICATIONS:

CARBOplatin

- **Hypersensitivity:** Reactions to CARBOplatin may develop in patients who have been previously exposed to platinum therapy. However allergic reactions have been observed upon initial exposure to CARBOplatin.
- **Neurotoxicity and ototoxicity**: Neurological evaluation and an assessment of hearing should be performed on a regular basis, especially in patients receiving high dose CARBOplatin. Neurotoxicity, such as parasthesia, decreased deep tendon reflexes, and ototoxicity are more likely seen in patients previously treated with cisplatin, other platinum treatments and other ototoxic agents. Frequency of neurologic toxicity is also increased in patients older than 65 years.

5-Fluorouracil

- **Myocardial ischaemia and angina:** Cardiotoxicity is a serious complication during treatment with fluorouracil. Patients, especially those with a prior history of cardiac disease or other risk factors, treated with 5-Fluorouracil, should be carefully monitored during therapy.
- **Dihydropyrimidine dehydrogenase (DPD) deficiency:** DPD is an enzyme encoded by the DPYD gene which is responsible for the breakdown of fluoropyrimidines. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidine-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. Treatment with 5-Fluorouracil, capecitabine or tegafur-containing medicinal products is contraindicated in patients with known complete DPD deficiency. Consider a reduced starting dose in patients with identified partial DPD deficiency. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring. Therapeutic drug monitoring (TDM) of fluorouracil may improve clinical outcomes in patients receiving continuous 5-Fluorouracil infusions.
- **Hand-foot syndrome (HFS):** HFS, also known as palmar-plantar erythrodysaesthesia (PPE) has been reported as an unusual complication of high dose bolus or protracted continuous therapy for 5-Fluorouracil.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Calais G, Alfonsi M, Bardet E, et al. Randomized trial of radiation therapy versus concomitant

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2. Appropriate chemotherapy dosing for obese adult patients with cancer: American Society of Clinical Oncology Clinical Practice Guideline. J Clin Oncol 2012; 30 (13) 1553-1561.
3. Szturz, P., Cristina, V., Herrera Gómez, R. G., Bourhis, J., Simon, C., & Vermorken, J. B. (2019). Cisplatin Eligibility Issues and Alternative Regimens in Locoregionally Advanced Head and Neck Cancer: Recommendations for Clinical Practice. Frontiers in Oncology, 9. https://doi.org/10.3389/fonc.2019.00464
4. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
5. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
6. HPRA Direct Healthcare Professional Communication. 5-Fluorouracil (i.v.), capecitabine and tegafur containing products: Pre-treatment testing to identify DPD-deficient patients at increased risk of severe toxicity. Accessed Aug 2020. Available at: https://www.hpra.ie/docs/default-source/default-document-library/important-safety-information-from-marketing-authorisation-holders-of-products-containing-5-fluorouracil-(i-v-)-capecitabine-and-tegafur-as-approved-by-the-hpra.pdf?sfvrsn=0
7. CARBOplatin 10mg/mL Summary of Product Characteristics. Last updated 18/07/25. Accessed January 2026.Available at: https://assets.hpra.ie/products/Human/27514/Licence_PA2315-080-001_04092024114146.pdf
8. Fluorouracil 50mg/mL injection Summary of Product Characteristics. Last updated 29/10/2025. Accessed Jan 2026. Available at: https://assets.hpra.ie/products/Human/27540/Licence_PA2315-091-001_05032025160439.pdf

From 16/09/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

Version	Date	Amendment	Approved By
1	18/09/2020		Prof Maccon Keane
2	22/10/2021	Reviewed. Updated drug interactions.	Prof Maccon Keane
3	14/12/2023	Updated baseline tests and exclusions sections. Updated renal and hepatic dose modifications table. Update to other supportive care section.	Prof Maccon Keane
3a	13/03/2025	Additional wording added to baseline testing section.	NCCP

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4	26/03/2026	Regimen reviewed. Updated ICD-10 codes.Updated regular tests. Updated renal and hepatic dose modifications table to align with Giraud et al 2023. Updated supportive care section. Updated regimen throughout in line with NCCP standardisation.	Dr Colm Mac Eochagáin
5	16/09/2026	Update to indication wording Update to eligibility section. Update to review schedule.	Dr Colm Mac Eochagáin

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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